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Acoustic evaluation of orchestra occupancies in concert halls: Effect of sound absorption by orchestra members on audience acoustics

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ABSTRACT

This study investigated changes in the reverberation time at concert hall seats due to the sound absorption of orchestra musicians. Different percentages of musician-occupied areas and densities of the musicians were analyzed on the stages of two 1:10 scale concert hall models. Increasing numbers of musicians was associated with a significant increase in the early-to-late sound index and a significant decrease in the reverberation time at the seats. The effects of volume and absorption coefficients associated with the presence of musicians were investigated in simple computer simulations of shoebox and fan-shaped halls. The results revealed that changes in the orchestra absorption ratio had a greater effect on the reverberation time at the seats than that of the total sound absorption. Therefore, for any given number of musicians in the scale model, the reverberation time at the seats decreased as the percentage of the stage area occupied by the orchestra increased. The reverberation time decreased by more than 0.5 s when the occupied area reached 90%. Furthermore, computer simulations of 12 concert halls showed that the reverberation time at the seats varied in relation to the orchestra absorption ratio.

1. Introduction

Concert hall acoustics would be ideally evaluated with musicians and audience members occupying the space in the same manner as during an actual performance. However, ISO 3382-1 requires that the seat occupancy status and measurement conditions be described in the test report because acoustic measurements are generally performed in an empty hall [1]. In fact, scale models are often used to investigate the sound absorption status around the sound source. In one such scale model experiment, Barron and Lee [2] found that the sound pressure at the seats was 1.3 dB less when the hall was empty than when it was fully seated. Furthermore, when Jeon and Barron [3] placed orchestra musicians on the stage in a scale model experiment, they found that the sound reflection from the rear wall of the stage was reduced by the musicians, but that the reduction was offset by the sound reflection from the bodies of the musicians. Dammerud and Barron investigated the sound propagation in an orchestra [4] and discovered that wider spacings between musicians increased the attenuation of high-frequency sounds. Beranek and Hidaka [5] calculated the total sound absorption (S_T) in a concert hall with musicians onstage; however, they neglected to address the effect of the sound absorption by musicians on the sound field of the concert hall. Studies on concert hall acoustics that consider the presence of musicians have been conducted using actual concert halls and scale models [6,7], as well as computer simulations [8,9].

The audience's perception of orchestral performances has also been researched. Galiana et al. [10] examined the differences in acoustic perceptions of experts and nonexperts in an actual concert hall, and Gimenez et al. [11] investigated the correlations between room acoustic indicators and the perceptions of audience members in the seats, again in an actual concert hall. Jeon et al. [12] studied the preferences of orchestral musicians depending on their placement on the stage; however, this study did not address the sound absorption effects on audience acoustics according to the musicians' stage occupancy.

The seats, audience, and musicians all commonly have high sound absorption factors, and the number of audience members in the seats has the largest effect on reducing reverberation in a concert hall. Initial studies on the sound absorption by audience members in concert halls were conducted by Beranek et al. who studied the effects of fully occupied seats [5,13,14]. To predict the reverberation time (RT) in the design of a concert hall, Beranek and Hidaka [5] classified upholstered chairs as having light, medium, or heavy upholstery, and regarded the audience as one block because of the uniform spacing and high density of individual members. Other studies [15–17] also investigated the correlations between laboratory measurements and predicted values. To quantify the sound absorption of seats and audiences, ISO 354 [18]

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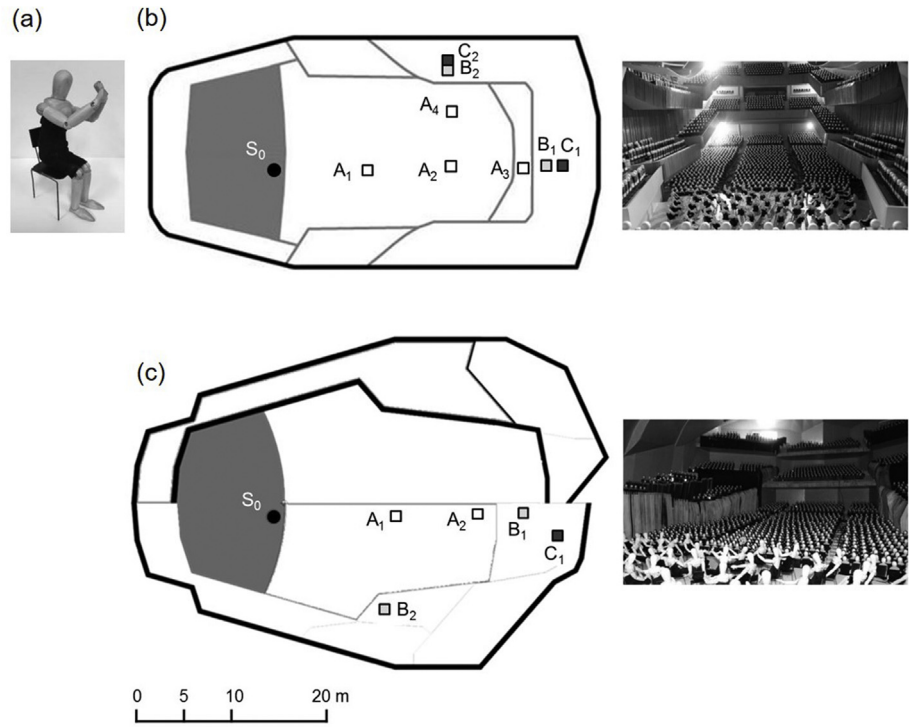


Fig. 1. (a) Configuration of the 1:10 scale model musician. (b) and (c) Floor plans of the 1:10-scale model halls with source (S_0) and receiver positions in (b) Hall A and (c) Hall B. A_n, B_n, and C_n indicate the receiver positions on the main floor, the first balcony, and the second balcony, respectively.

assigns an equivalent sound absorption area of the seats based on the number of seats. However, to calculate a room's acoustic parameters by computer simulation, the sound absorption area of audience seats must be converted to an absorption coefficient for the area of the floor that is occupied by chairs. These absorption coefficients, which are measured with various chair arrangements in the reverberation chamber and used to calculate the absorption coefficients of audience seats, have shown a linear relationship with the actual perimeter/area (P/A) [19]. In addition, the actual measured RT value has been found to match well with the absorption coefficient in a concert hall because the RT value reflects the difference in the total volume of the concert hall due to the sound absorption of the seats, which is predicted by considering the audience as a single seating block [20,21]. However, the P/A method requires repeated measurements to predict the various sound absorption amounts associated with the distinct P/A ratios of different samples. Therefore, to determine the absorption coefficient of an orchestra, the absorption coefficients of seats in a reverberation chamber must be measured for different arrangements, because although the absorption coefficient per seat is constant, the absorption coefficient is also affected by the spacing between seats [16,22,23].

However, musicians and audience members present different sound absorption characteristics, such as seating arrangement, number of exposed edges, and distance from the sound source. Just as the sound absorption of the seats occupied by the audience is influenced by the absorption coefficient of the chairs, the effect on sound absorption of the onstage musicians varies in relation to the size of the empty stage, which constitutes a reflective floor. According to Barron [24], an orchestra of 100 musicians, including percussion instruments, occupies an area of approximately 150 m^2. Therefore, an almost 200-m^2 stage is required to accommodate solo musicians and percussion instruments, as well as a 1-m-wide empty space in front of the stage and additional spaces for the stage riser and access ways. Therefore, to investigate the effects of musicians on concert hall acoustics, the absorption coefficient of the musicians who occupy most of the stage area must be included in the calculations.

In this study, to evaluate the acoustics of a concert hall that is fully occupied by both musicians and audience members, the absorption coefficients associated with the musicians were measured with various musician arrangements in a reverberation chamber, using two 1:10 scale concert hall models [25,26] and 10 computer simulations of actual concert halls. The effects of the sound absorption of musicians on acoustic parameters in the audience were investigated according to various sound absorption characteristics, space volumes, and numbers of musicians, and the measurement data from the two experimental models were also applied in additional computer simulations.

2. Scale model

In architectural acoustic design, 1:25 or higher scale models have been used to evaluate the acoustic effects of complex shape factors such as organs or diffusers in concert halls [27,28]. In this study, the absorption coefficient of musicians fabricated as 1:10 scale models were first measured in a reverberation chamber, and then the sound absorption effects of the presence or absence of musicians on the acoustic indicators at the concert hall seats were investigated using a 1:10 scale model of a concert hall.

2.1. Model musician and hall

The absorption coefficient of orchestra musicians was measured in accordance with ISO 354 [18]. First, to consider the number of orchestra musicians and the types of instruments on the concert hall stage, the average stage area per musician was set at 1.9 m^2 based on Beranek [29], and for the sound absorption surface area per cello musician, the equivalent sound absorption area was set at 2.3 m^2/person based on Harwood and Burd [30]. The musician model was constructed based on these values as shown in Fig. 1(a); extrapolated to life size, this model would have a waist measurement of 0.35 m, a shoulder measurement of 0.55 m, and height of 1.35 m when seated. The bodies of the 1:10 scale musicians were fabricated of wood, and their chests and upper legs were wrapped with 1-mm-thick fabric. The chairs of the musicians were made of polyurethane plastic sheets and iron wires. The

average absorption coefficient of the 1:10 scale musician chair models was low at 0.06 in the 100–5000 Hz band [22]. With respect to the areas of the instruments, Barron [24] suggested a minimum surface area of 1.25 m^2 for violin and wind instruments, 1.5 m^2 for cello and large wind instruments, 1.8 m^2 for double bass, 10 m^2 for timpani, and 20 m^2 for other percussion instruments.

To evaluate the effects of the musician density onstage on the sound absorption of the audience, this study investigated various musician densities based on a single musician occupying 1.5, 1.8, 2.0, or 2.3 m^2 of the stage floor area, excluding large percussion. As shown in Fig. 1 (b) and (c), 1:10 scale models of a shoebox-shaped hall (Hall A) with 2100 seats (stage area: 193 m^2) and a fan-shaped hall (Hall B) with 1750 seats (stage area: 215 m^2) were used to investigate the changes in the audience acoustic indicators due to the sound absorption of musicians.

2.2. Measurement

Eight measurement points were selected in Hall A and five in Hall B, as shown in Fig. 1 (b) and (c); measurement points in each hall were selected to take one sample point for approximately every 300 seats. The number of musicians was then changed, with their numbers determined by the stage size and orchestra composition (Hall A: 0, 48, 67, 89, or 98 musicians; Hall B: 0, 37, 46, 66, or 90 musicians). With the increasing number of musicians, the musician density increased, with each musician occupying 2.3 m^2 originally versus a final 1.4 m^2 in Hall A and an original 2.1 m^2/person versus a final 1.5 m^2/person in Hall B.

3. Results

Fig. 2 shows the sound absorption amount per person in each frequency band measured in the reverberation chamber using actual persons and musician models [22,30]. The model results were similar for the four musician densities, and the results showed similar characteristics to those reported by Harwood and Burd [30] for frequencies above 1 kHz. The sound absorption amount of the musician models in the low frequency bands below 500 Hz was between the sound absorption amounts measured from actual cellists and actual musicians [22,30], and realistic values were shown by the panel sound absorption effect of the instruments in the 125 Hz band [4]. As shown in Fig. 2, the amount of sound absorbed by each musician model did not change significantly even when the density changed from 1.5 to 2.3 m^2/person. This result suggests that the sound absorption amount per musician depends on the absorption coefficient of the musicians (specimens)

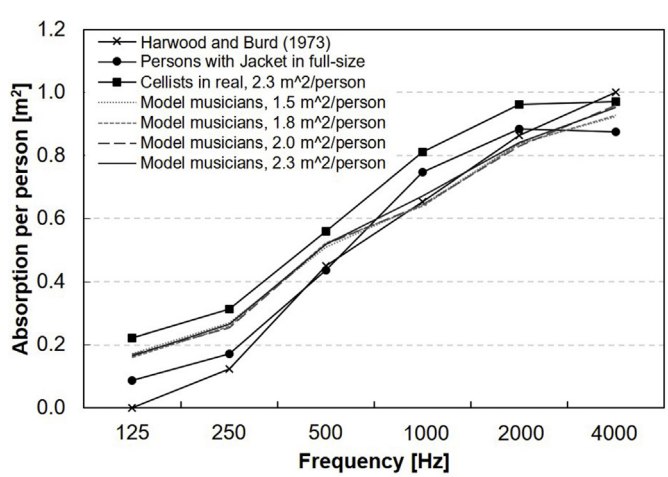


Fig. 2. Real measurements of absorption per musician in a reverberation chamber, from Harwood and Burd [30], and measured with actual cellists (Jang and Jeon [22]). The absorption coefficients for the model musicians with various seating densities matched the actual measurements.

themselves, not on the spacing between musicians. Therefore, the density of musicians was considered irrelevant to the total absorption coefficient of the orchestra, and the greatest factor affecting the stage sound absorption was the area occupied by the orchestra.

Next, the effect of the sound absorption of the orchestra on the entire hall was evaluated, and changes in RT, early decay time (EDT), sound strength (G), and early-to-late sound index (C80) were observed when the number of musicians was increased in scale models of Hall A and Hall B with fully occupied audience seats as shown in Fig. 3. The RT and EDT showed relatively distinct linear relationships with the number of musicians. When the stage was empty, the RT value in the middle frequency band was 2.04 s in Hall A and 2.11 s in Hall B. When there were 90 musicians, the RT value decreased by 0.10 s in Hall A and by 0.19 s in Hall B, on average, compared with the empty stage. The EDT value when the stage was empty was 1.93 s in Hall A and 1.97 s in Hall B. In comparison, when there were 90 musicians, the EDT value decreased by 0.17 s in Hall A and by 0.21 s in Hall B, and the G value decreased by 0.5 dB and 0.3 dB in Hall A and Hall B, respectively. Furthermore, the C80 value increased by 0.8 dB and 0.4 dB, respectively.

As described above, distinct patterns of change in the acoustics at the audience seats emerged in response to the changing amount of sound absorbed by onstage musicians, but the change relationships varied based on the shape, volume, surface absorption coefficient, size, and material finish of the concert hall. Furthermore, as described above, the two halls showed some differences in the change amount depending on the musician seating arrangement and the number of musicians; nonetheless as shown in Fig. 3, Hall A and Hall B showed similar general trends in the changes of acoustic indicators according to the number of musicians.

4. Computer simulation

To apply the relationships shown by the experimental models to the acoustic design of actual halls, they should be generalized by studying the effects of the number of musicians in actual concert halls of various shapes. In this study, the stage area occupied by the musicians was investigated by computer simulations using the Odeon version 13 software application for simple shoebox- and fan-shaped models [26], such as those adopted for Hall A and Hall B in the scale model experiments, as well as the 3D models of 10 existing halls. In the Odeon simulations, the transition order (i.e., change in the reflection order associated with a change from the early-image source method to the late-radiosity method) was set to be second order. To calculate the concert hall's acoustic indicators, the impulse response length was set at 3500 m s, and 5000 late rays were used, which is slightly higher than the recommended value to enable comparison between different models. In Equation (1), the orchestra absorption ratio (OAR) is represented as the amount of sound absorbed by the orchestra musicians divided by the amount of sound absorbed by the entire concert hall, and this ratio is used to compare the effects of the musicians' sound absorption in different spaces.

$$OAR(Orchestra\ Absorption\ Ratio) = S_{orc}a_{orc} / \sum_{i}^{n} s_i a_i \times 100[\%] \quad (1)$$

S_{orc}: Surface area of orchestra
α_{orc}: Orchestra absorption coefficient
S_i: Area of each material
α_i: Absorption coefficient of each material
n: Number of materials

4.1. Simple models

First, to examine the effects of the sound absorption of orchestra musicians on the concert hall's acoustic indicators, two types of simple

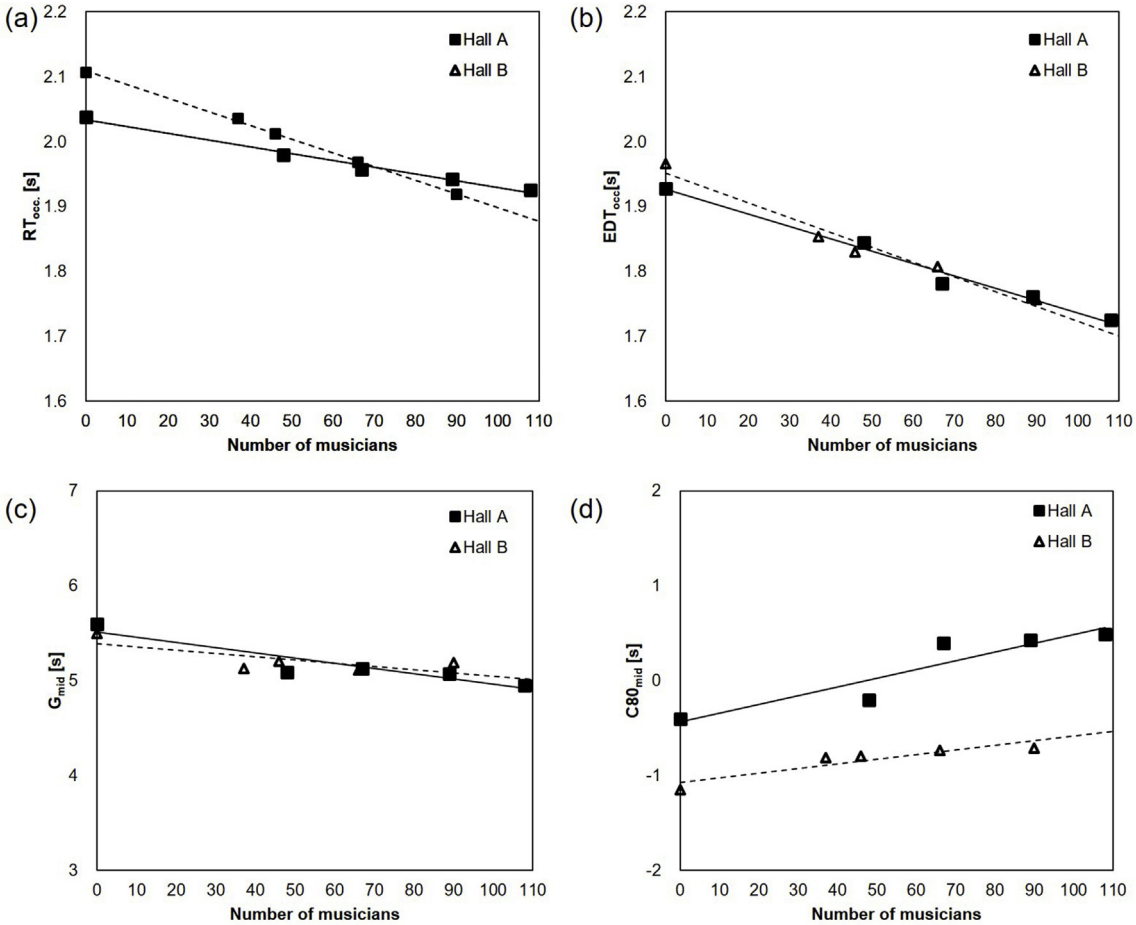


Fig. 3. Measured acoustic parameters versus the number of musicians in Hall A and Hall B: (a) reverberation time (RT), (b) early decay time (EDT), (c) sound strength (G), and (d) d-to-late sound index (C80).

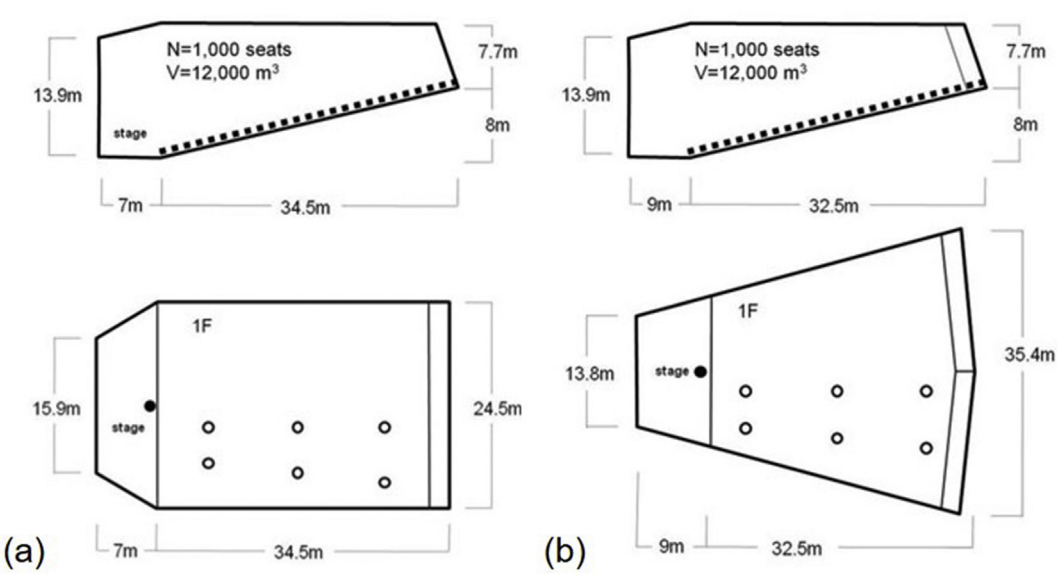
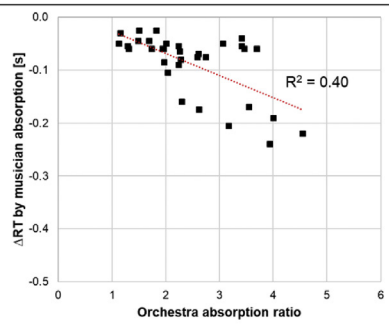
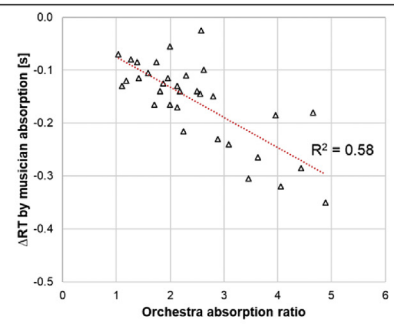


Fig. 4. Simple reference models of shoebox- and fan-shaped halls used to evaluate the influence of orchestra occupancy in computer simulations [26]. Section and floor plan of (a) the fan-shaped hall and (b) the shoebox-shaped hall.

models (reference halls) with different volumes were evaluated. The reference concert halls were shoebox- and fan-shaped halls with similar sizes, including a volume of 12,000 m^3 and 1000 seats, even though their shapes were different, as shown in Fig. 4 [26]. The volumes of the concert halls were symmetrically increased in 2-m units in four stages in each of the *x*-, *y*-, and *z*-directions, such that the maximum volumes of the two concert halls were 26,000 m^3 and 24,500 m^3, respectively. The numbers of musicians for the reference halls were set at 40, 70, and 90. The absorption coefficients of the seat chairs (reference models had "medium upholstered chairs") and walls were adjusted so that the RTs

Table 1
Simple models of fan- and shoebox-shaped halls used for the evaluation of orchestra absorption effects.

Hall	Shoebox			Fan		
Model						
RT [s]	V [m^3]	OAR [%]	ΔRT [s]	V [m^3]	OAR [%]	ΔRT [s]
1.0	12,000	3.4–3.7	0.04–0.06	12,000	1.7–2.8	0.09–0.15
	14,000–24,500	1.1–2.6	0.03–0.09	15,500–26,000	1.0–2.6	0.09–0.17
2.0	12,000	2.6–4.6	0.18–0.24	12,000	2.9–4.9	0.23–0.35
	15,000–24,500	1.8–4.0	0.02–0.21	16,000–26,000	2.0–4.7	0.03–0.32

of the reference models would become 1.0 s and 2.0 s, respectively, when the volumes of these two halls with different shapes were changed.

When the total volume was changed from 14,000 m^3 to 24,500 m^3 in the shoebox-shaped hall as shown in Table 1, the total sound absorption amount (total absorption) increased from 1800 m^2 to 2300 m^2, and the OAR decreased by at least half; furthermore, the RT changed substantially from 0.02 s to 0.21 s. In the fan-shaped hall, when the volume was changed from 15,500 m^3 to 26,000 m^3, the OAR decreased at least by a half and the reverberation time changed from 0.03 to 0.32. In particular, both halls showed a wider range of reverberation variations by orchestra musicians when the RT was 1 s compared with a reverberation time of 2 s.

The results of regression analysis on the number of orchestra musicians, OAR, total acoustic absorption, and RT showed that the total R^2, at 0.384, was statistically significant ($F(3,74) = 15.367$, $p < 0.001$) and predicted or explained a portion of the reverberation changes in response to the number of musicians, OAR, and total sound absorption. Furthermore, in the regression coefficient analysis, only the OAR regression coefficient ($b = 0.048$, $p < 0.001$) was significant in both the shoebox- and fan-shaped halls. As shown in Table 1, the influence of the OAR in the fan-shaped hall was especially clear ($R^2 = 0.58$). Consequently, the effect of stage occupancy by orchestra musicians changed most when the OAR ratio increased and had the greatest effect on the decreasing RT at the seats.

4.2. Twelve halls

As shown in Table 2, the computer simulation results were compared for a total of 12 concert halls, including 10 existing concert hall models in the Odeon library and the above-mentioned two 1:10 scale concert halls (Hall A and Hall B), with the stage occupied by a large orchestra. The 12 concert halls included four fan-shaped halls, five shoebox-shaped halls, and three additional halls (London Barbican, Cardiff St. Davis Hall, and Edinburgh Usher Hall), with over 1000 seats and various shapes. For the Odeon library models, the surface absorption coefficient was applied based on the measurement data from the unoccupied concert halls [31,32]. In this study, however, the RTs of the halls were adjusted to make them similar to the actual measurement data with the stage and seats fully occupied by musicians and audience members [33]. With respect to the chairs of the concert halls in the simulation, the absorption coefficients of occupied highly-, intermediately-, and lightly-upholstered chairs proposed by Beranek and Hidaka [5] were selectively applied in line with each hall. The various data from the scale model of Hall A and Hall B, including OAR and RT, were used for the simulations.

4.3. Simulation

To evaluate the effects of orchestra musicians through simulations, measurements from the scale models must be used to determine the effects of absorption coefficient and for computational modeling [22]. For this purpose, in this experiment, the equivalent sound absorption area per person (A_{obj}) was measured in a reverberation chamber in each frequency band and converted to the absorption coefficient (α_s) before the RT was determined by computer simulation. Musicians were then placed in the simulated concert halls, and the changes in concert hall acoustics according to the number of musicians were predicted and compared with the results of the 1:10 scale models.

In the 12 concert halls, 90 musicians were arranged uniformly at a density of 1.5 m^2/person. The equivalent sound absorption area (A_{orc}) of the 90 orchestra musicians was 52.2 m^2 in the middle frequency band, which was calculated by multiplying the musician sound absorption amount per person (1.5 m^2/person, 0.52 in Fig. 3) by the number of musicians (90). This result is similar to the equivalent sound absorption area of 52 m^2 estimated for 92 orchestra musicians in the Tokyo Opera City Concert Hall (TOC) [5]. The RTs were calculated at 20 points in the audience seating arrangement (existing halls) with and without orchestra musicians.

Table 3 lists the measured sound absorption amounts per musician under various seating densities converted to absorption coefficients (α). These absorption coefficients were highest at the high musician density (1.5 m^2 per person) because the same sound absorption amount was divided by a smaller unit of occupied floor area. In other words, the absorption coefficient divided by the absorption coefficient per unit area (α_c) was higher than that divided by the absorption coefficient per unit of occupied area (α_O).

Furthermore, as shown in Table 4, the absorption coefficient varied with the density and perimeter (P) of musicians, and the area of specimens (S or A) [18]. The absorption coefficient for the orchestra

Table 2
Data used for the evaluation of the orchestra absorption in seven existing concert halls when occupied.

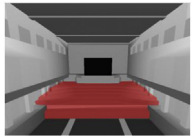
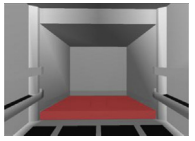
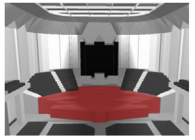
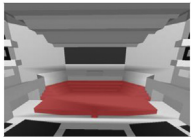
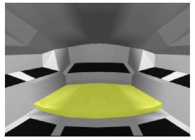
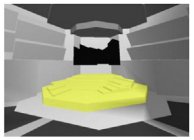
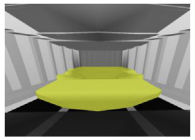
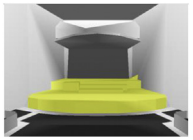
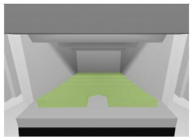
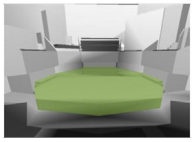
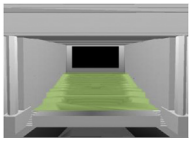
Hall	Hall A (HA)	Grosser Musikvereinssaal, Vienna (VM)	Symphony Hall, Boston (BO)	Concertgebouw, Amsterdam (AM)	Royal Festival Hall, London (LR)	Hall B (HB)
Stage view						
Shape	Shoebox	Shoebox	Shoebox	Shoebox	Shoebox	Fan
N	2100	2158	2625	2037	2901	1750
V [m^3]	19,600	15,000	18,750	18,780	22,000	18,500
S_A [m^2]	1048	955	1370	1125	1972	1044
S_O [m^2]	214	185	150	160	225	193
Hall	**Philharmonic am Gasteig, Munich (MG)**	**Festspielhaus, Salzburg (SF)**	**Konserthus, Gothenberg (GK)**	**Barbican, London (LB)**	**St. Davis Hall, Cardiff (CW)**	**Usher Hall, Edinburgh (EU)**
Stage view						
Shape	Fan	Fan	Fan	Other	Other	Other
N	2387	2158	1286	2026	1952	2547
V [m^3]	29,737	15,500	11,900	17,750	22,000	15,700
S_A [m^2]	1639	1375	666	1265	1235	1204
S_O [m^2]	231	204	170	217	195	154

Table 3
Absorption coefficients of model musicians as a function of musician density (from 1.5 to 2.3 m^2/person). P/A ratio of the orchestra, β (the slope of the regression line), and α_∞ (the intercept of the regression line) were used to obtain the absorption coefficients.

Density [m^2/person]		Frequency [Hz]					
		125	250	500	1 k	2 k	4 k
1.5	β	0.03	0.04	0.09	0.17	0.13	0.12
	α_∞	0.05	0.08	0.14	0.24	0.31	0.38
	α	0.06	0.10	0.18	0.31	0.36	0.43
1.8	β	0.04	0.02	0.08	0.17	0.12	0.13
	α_∞	0.04	0.09	0.13	0.18	0.35	0.36
	α	0.06	0.10	0.16	0.25	0.40	0.41
2.0	β	0.01	0.01	0.08	0.12	0.1	0.12
	α_∞	0.05	0.08	0.11	0.21	0.33	0.35
	α	0.05	0.08	0.14	0.26	0.37	0.40
2.3	β	0.03	0	0.08	0.14	0.13	0.17
	α_∞	0.02	0.09	0.1	0.17	0.25	0.24
	α	0.03	0.09	0.13	0.23	0.30	0.31

Table 4
Influence of orchestra density on the RT: (a) fixed occupied area, (b) fixed number of musicians.

	Area per musician [m^2/person]	1.5	1.8	2.0	2.3
(a) Same occupied area	Area [m^2]	133			
	Musicians	89	75	67	58
	ΔRT (HA) [s]	−0.13	−0.12	−0.12	−0.11
	ΔRT (HB) [s]	−0.17	−0.15	−0.15	−0.14
(b) Same number of musicians	Area [m^2]	113	133	150	173
	Musicians	75			
	ΔRT (HA) [s]	−0.08	−0.12	−0.12	−0.14
	ΔRT (HB) [s]	−0.13	−0.15	−0.17	−0.17

seating density was determined from the equation of P/A ratio (0.41) with the regression coefficient (β) and fragment value (α_∞), and compared with other absorption coefficients.

The orchestra musicians were modeled with the net covered area (box shape). The height of the orchestra musician was set at 1.35 m, considering the height of sitting people. This modeling method has been suggested in a previous study [8]; however, certain model factors, such as the surface dimensions of the model, have not yet been detailed. In this study, the modeling results were calculated to considering the actual fully projected position of one orchestra musician. For example, if the area per musician was 2.3 m^2, the net area covered by musicians, including their spacing, was 20% of the total floor area. Therefore, the transparency coefficient was set at 0.8. The effects of input variable variations, such as those in the absorption and scattering coefficients, must be considered as well, and the scattering coefficient of the musician model was set at 0.4 [22].

4.4. Results

Fig. 5 shows the changes in RT with and without musicians in computer simulations and scale models in each frequency band. In Hall A, the RT decreased by up to 0.12 s, and the amount of decrease in the RT increased for higher frequencies. Hall B showed the largest amount of decrease in the middle frequency bands of 500 and 1000 Hz. When the computer simulation and scale model methods were compared with changing numbers and densities of musicians in a fixed arrangement, both halls showed similarly decreasing trends in RT as the number of musicians increased in the middle frequency bands (see Fig. 6).

In addition, the influence of the RT at audience seats was determined as a function of the orchestra density in Hall A and Hall B using computer simulations. As shown in Table 4, the absorption coefficient changed more in response to changes in the total area occupied by the orchestra, even when the number of orchestra musicians was fixed, than in response to changing numbers of musicians with a fixed total area. The RT decrease in Hall B was greater than that in Hall A because the RT value of Hall B was larger than that of Hall A when the stage was empty, with no musicians. Table 4 (b) shows the results when the occupied area was changed with a fixed number of orchestra musicians. The same number of orchestra musicians were arranged in different spacings, and the RT decrease was smaller when the orchestra was arranged at a higher density. Therefore, because the area occupied by musicians had a significant effect on the RT decrease at the seats, reducing the area occupied by musicians by minimizing the spacings between them can effectively decrease the amount of sound they absorb.

5. Discussion and conclusions

To investigate the changes in RT at the audience seats induced by the sound absorption of musicians, the OARs of 12 concert halls, including Hall A and Hall B, were calculated and are presented in Fig. 7. As shown in this figure, the RT decrease induced by the musicians was affected by the total sound absorption of the concert hall. The maximum RT decrease was 0.16 s, which was observed in Grosser Musikvereinssaal, Vienna, a hall with a relatively low total sound absorption. Concert halls with relatively low reverberations and high sound absorption areas, such as MG (Gasteig Philharmonie, Munich) and LR (Royal Festival Hall, London) showed almost no changes in

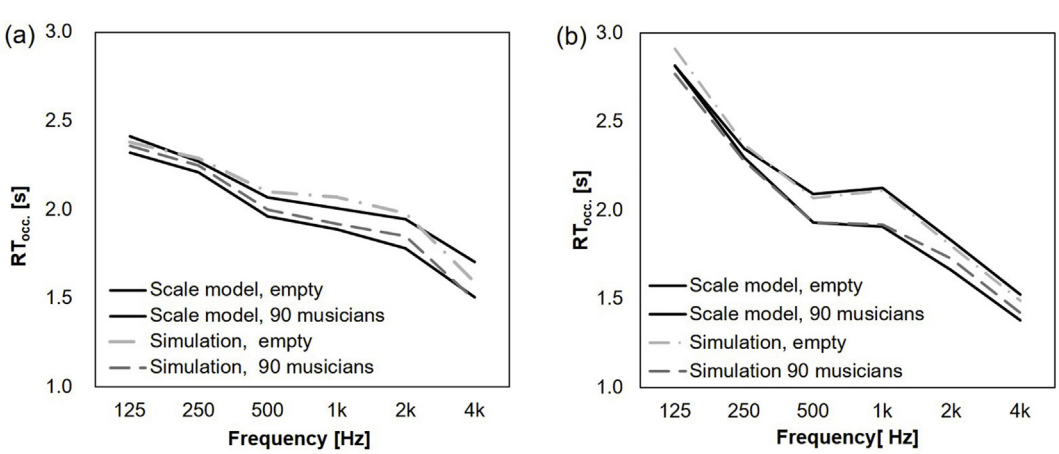


Fig. 5. Comparison of the RTs in the frequency domain for the scale models (solid lines) and computer simulations (dashed lines) for (a) Hall A and (b) Hall B, with a 90-piece orchestra.

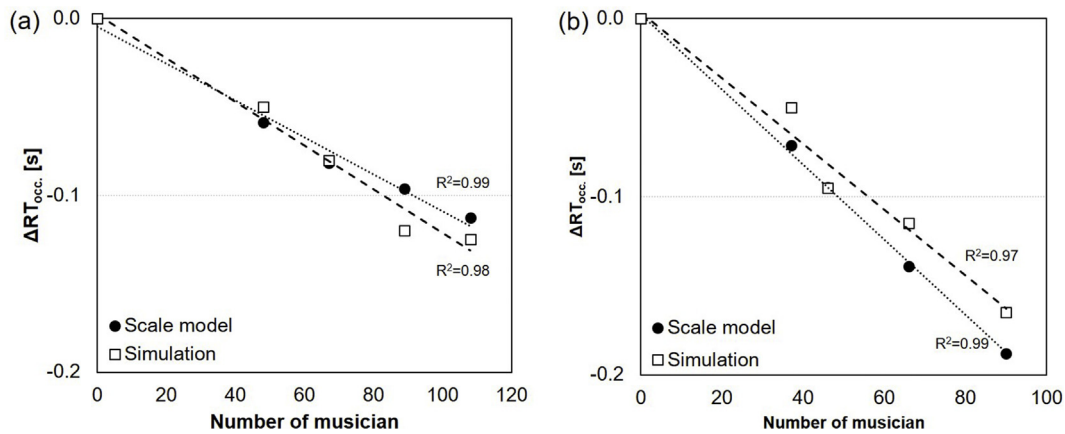


Fig. 6. Comparison of the RTs for the scale models and computer simulations for (a) Hall A and (b) Hall B, with different numbers of musicians.

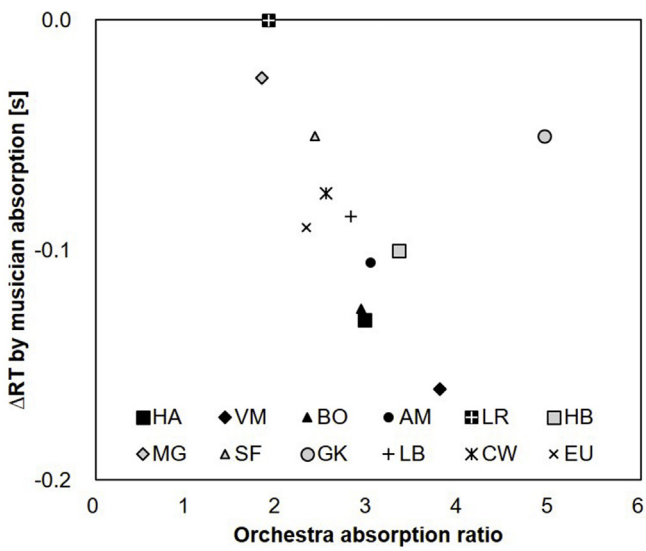


Fig. 7. RT difference versus orchestra absorption ratio in concert halls. (Black: shoebox-shaped halls; Gray: fan-shaped halls; Lines: Other shapes).

reverberations at the seats in response to OAR values.

This study investigated the effects of the number and density of musicians on the acoustic parameters at the audience seats using scale models and computer simulations. In the scale model experiments, the quantitative changes of RT were measured according to the density of orchestra musicians and the equivalent sound absorption areas of the internal components of the concert halls. The results of 1:10 scale model evaluations with orchestra musicians showed that the RT, EDT, and G decreased and C80 increased as the number of musicians on the stage increased. When 90 musicians occupied the stage, RT and EDT decreased by 0.16 s and 0.21 s, respectively, but G decreased by 0.5 dB and C80 increased by approximately 1 dB. Furthermore, G and C80 were more affected by the density rather than the number of musicians on the same stage.

The computer simulations of the 12 concert halls were conducted to examine the sound absorption effects of musicians on various shapes of concert hall stages, and the simulations revealed that the acoustic parameters changed according to changes in the density of musicians. This result suggests that the actual area occupied by musicians has a greater effect on the decrease of RT at the audience seats than the number of musicians on the stage. In particular, smaller spacings between orchestra musicians had a smaller effect on absorption, resulting in a smaller RT decrease. Furthermore, the decrease of RT induced by the musician density was affected by the overall sound absorption area of the concert hall, such that the musician density had a greater effect in concert halls with a longer RT. In future research, the influence of orchestra musicians on sound absorption can be practically verified by comparing the results derived in this study with measurement results in real halls.

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